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SchoolofComputerScienceandArtificialIntelligence

LabAssignment#9.2

Program	**:B.Tech(CSE)**
Specialization	**:**
Course Title	**:AIAssistedCoding**
CourseCode	**:23CS002PC304**
Semester	**II**
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Name of Student	**:J harshith yadav**
Enrollment No.	**: 2403A51L49**
Batch No	**52**
Date	**:06/02/26**

TaskDescription-1(Documentation–FunctionSummaryGeneration)

Codeused:

```
def factorial(n):
  ifn == 0:
    return1
  else:
    returnn*factorial(n-1)
factorial(5)
```

Prompt: Addashortsummarydescribingthepurposeofthefunctioninthecode

Output:

```
[1]
def factorial(n):
  # This function calculates the factorial of a non-negative integer using recursion.
  if n == 0:
    return 1
  else:
    return n*factorial(n-1)

factorial(5)

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```

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TaskDescription-2(Documentation–LogicalExplanationfor Conditions and Loops)

CodeUsed:

```
defnumbers(n):
    if n > 0:
        result="Positive"elif
    n < 0:
        result="Negative"
    else:
        result="Zero"return
    result
numbers(3)
```

Prompt:explainonlydecision-makinglogicandloopbehaviorinsidethecode

Output:

```
[7]
def numbers(n):
    # Decision-making logic: Determines if 'n' is positive, negative, or zero.
    # No loops are present in this function; execution flows through conditional branches.
    # Check if 'n' is positive
    if n > 0:
        result = "Positive"
    # If not positive, check if 'n' is negative
    elif n < 0:
        result = "Negative"
    # If neither positive nor negative, 'n' must be zero
    else:
        result = "Zero"
    return result
numbers(3)

... 'Positive'
```

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TaskDescription-3(Documentation–File-LevelOverview)

CodeUsed:

```
deffactorial(n):
    result = 1
    foriinrange(1,n+1):
        result *= i
    returnresult

def square_list(numbers):
    return [n*n forn in numbers]

def main():
    print(factorial(5))
    print(square_list([1,2,3,4])) main()
```

Prompt: Generateahigh-leveloverviewsummarizingtheoverallpurposeandfunctionality of the Code. add the summary at the beginning

Output:

```
[8]
0s
# This code defines two utility functions: `factorial` to calculate the factorial of a number,
# and `square_list` to square each element in a list. It then demonstrates their usage in a `main` function.

def factorial(n):
    result = 1
    for i in range(1, n + 1):
        result *= i
    return result

def square_list(numbers):
    return [n * n for n in numbers]

def main():
    print(factorial(5))
    print(square_list([1, 2, 3, 4]))

main()

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[1, 4, 9, 16]
```

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TaskDescription-4(Documentation–RefineExistingDocumentation)

CodeUsed:

```
def calculate_discount(price, discount):
    # calculate
    ifprice >0:
        # applydiscount
        final_price=price-(price*discount/100) return
        final_price
    else:
        #wrongprice return
        0
```

Prompt: Rewritethedocumentationtoimproveclarityandconsistency.

Output:

```
[10]
✓ 0s
def calculate_discount(price, discount):
    """
    Calculates the final price after applying a discount.

    Args:
        price (float or int): The original price of the item.
        discount (float or int): The discount percentage to apply (e.g., 10 for 10%).

    Returns:
        float: The final price after applying the discount. Returns 0 if the original price is not positive.
    """
    if price > 0:
        # Apply the discount percentage to the original price
        final_price = price - (price * discount / 100)
        return final_price
    else:
        # Return 0 if the price is invalid (non-positive)
        return 0
```

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TaskDescription-5(Documentation–PromptDetailImpactStudy)

CodeUsed:

```
defreverse_string(text):
    return text[::-1]
```

Prompt-1: Writeashortdocumentationcommentexplainingwhatthis Pythonfunction does.

Output-1:

```
[12]
✓ 0s
def reverse_string(text):
    """
    Reverses a given string.

    Args:
        text (str): The input string to be reversed.

    Returns:
        str: The reversed string.
    """
    return text[::-1]
```

Prompt-2:Writeaclearanddetaileddocumentationcommentforthegiven Python function.

Output-2:

```
[13]
✓ 0s
def reverse_string(text):
    """
    Reverses the given input string.

    This function takes a string as input and returns a new string
    with the characters in reverse order. It utilizes Python's string slicing
    feature to achieve this efficiently.

    Args:
        text (str): The input string that needs to be reversed.

    Returns:
        str: A new string containing the reversed characters of the input string.

    Examples:
        >>> reverse_string("hello")
        'olleh'
        >>> reverse_string("Python")
        'nohtyP'
        >>> reverse_string("12345")
        '54321'
    """
    return text[::-1]
```